

Lab 15

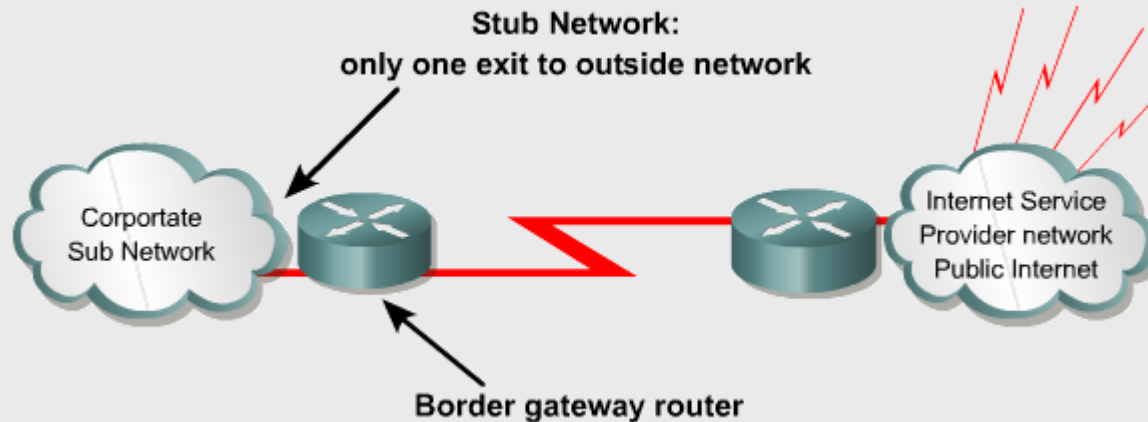
Network Address Translation

Introduction

- IPv4 addressing scalability issue
 - Internet growth
 - Static hierarchical addressing structure is inefficient

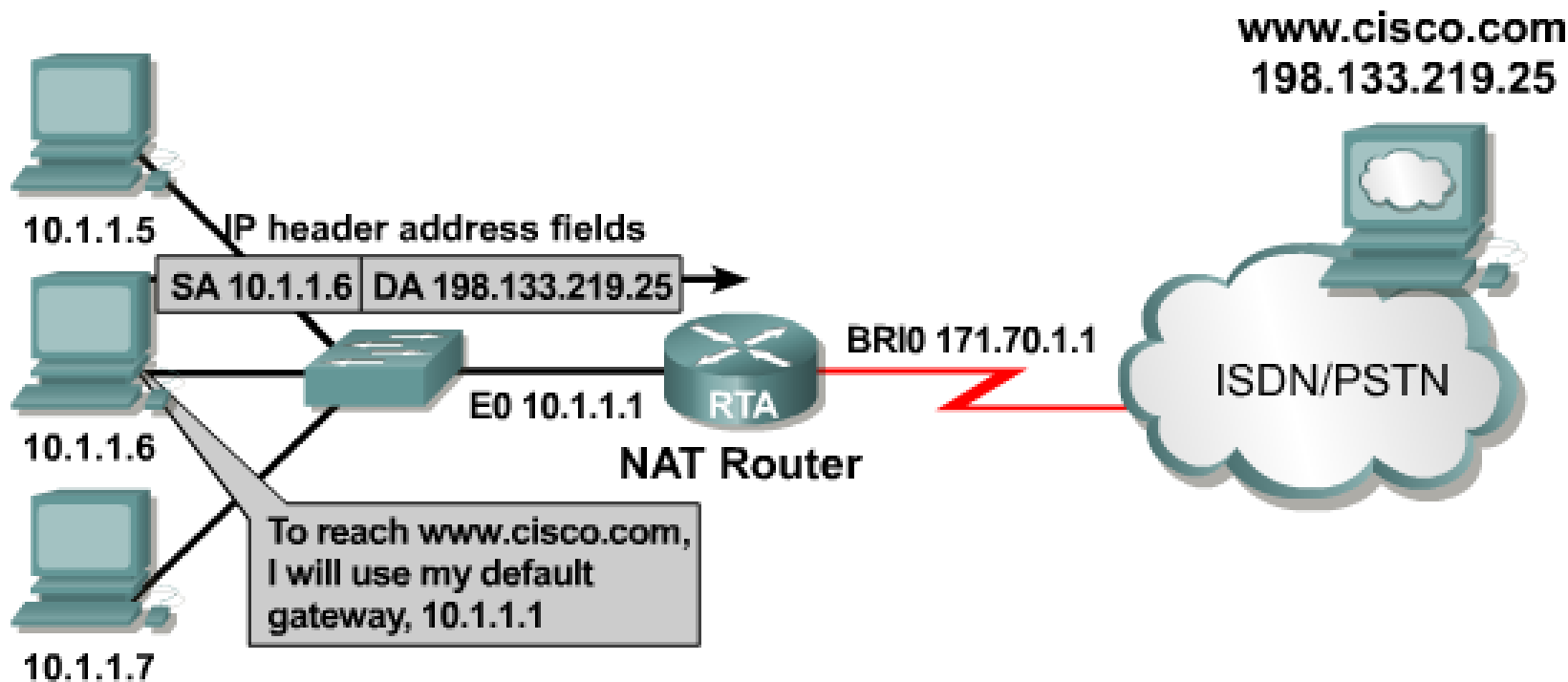
- Solutions
 - New version: **IPv6**
 - 128 bit addresses
(340.282.366.920.938.463.463.374.607.431.768.211.456 addresses)
 - **Variable Length Subnet Mask**: adapts a network address space to its real size
 - **Network Address Translation (NAT) [RFC 3022]**
 - Seamless substitution of IP addresses while packet is forwarded
 - Artificially increases the address space by allowing address reuse in different networks

Basic mechanism

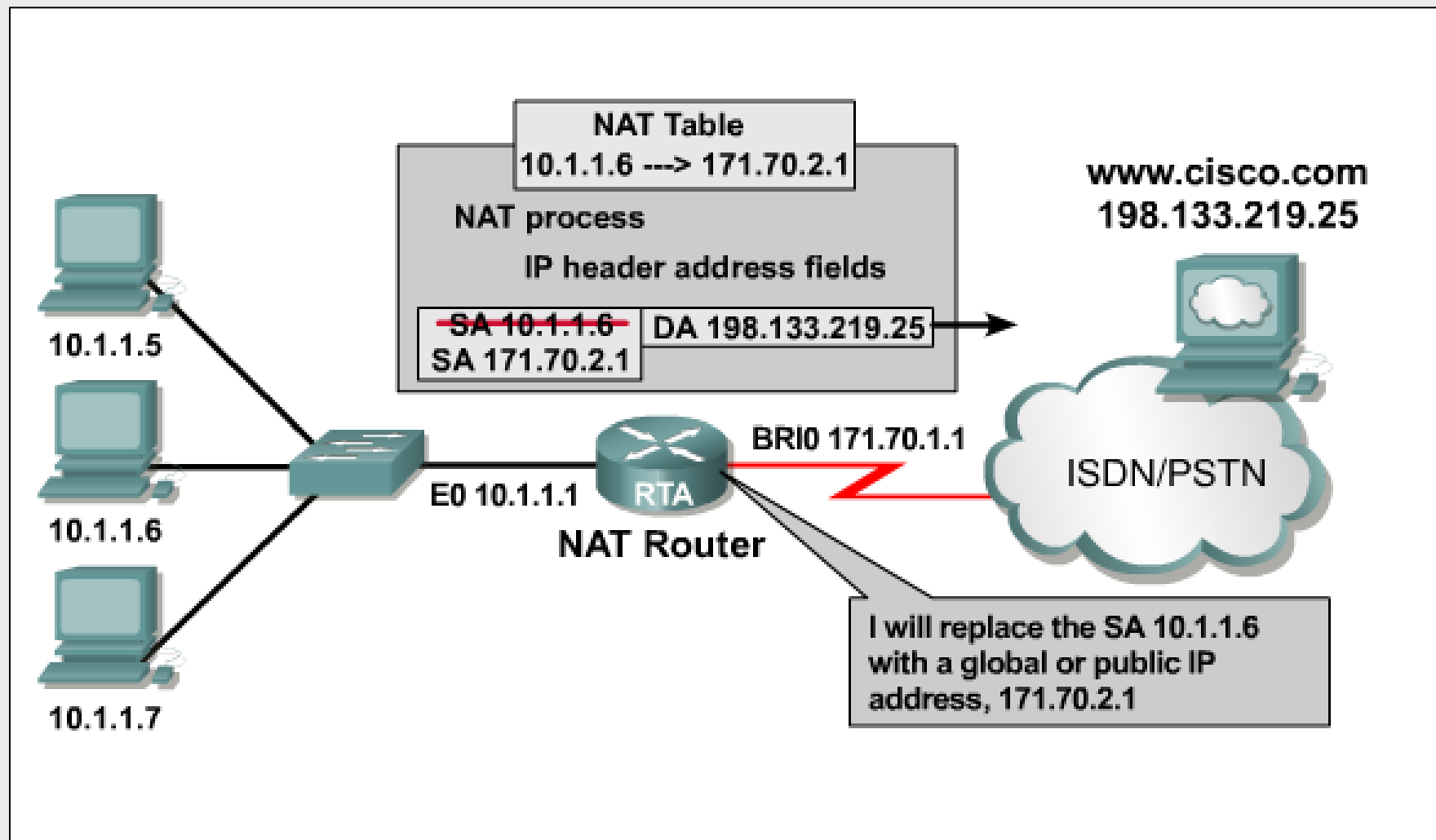


- Multiple *address realms*
- Two possible approaches (under *Traditional NAT*)
 - **Basic Network Address Translation or Basic NAT** – "is a method by which IP addresses are mapped from one group to another, transparent to end users."
 - **Network Address Port Translation, or NAPT** – "is a method by which many network addresses and their TCP/UDP ports are translated into a single network address and its TCP/UDP ports."
- Typically used on the border of a *stub network*

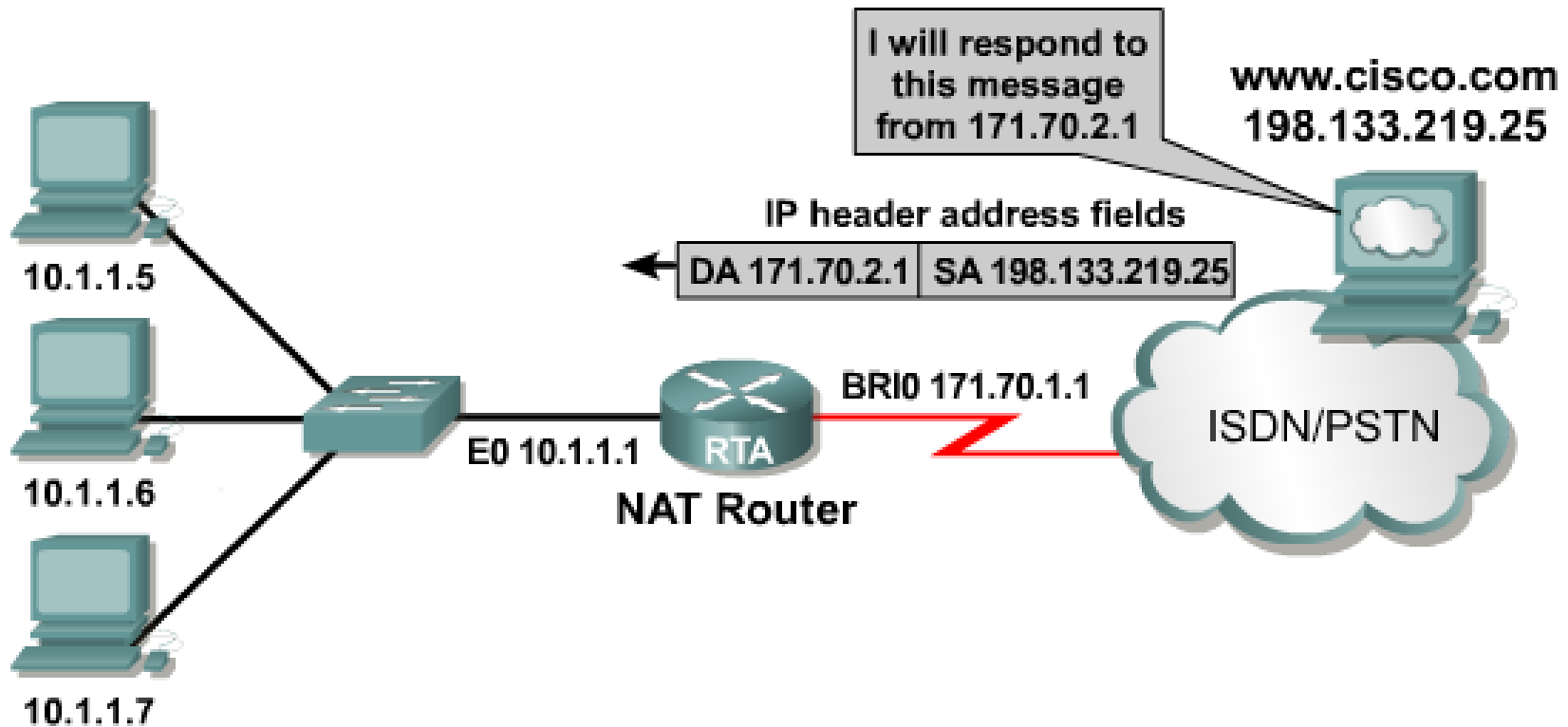
Basic NAT – Example



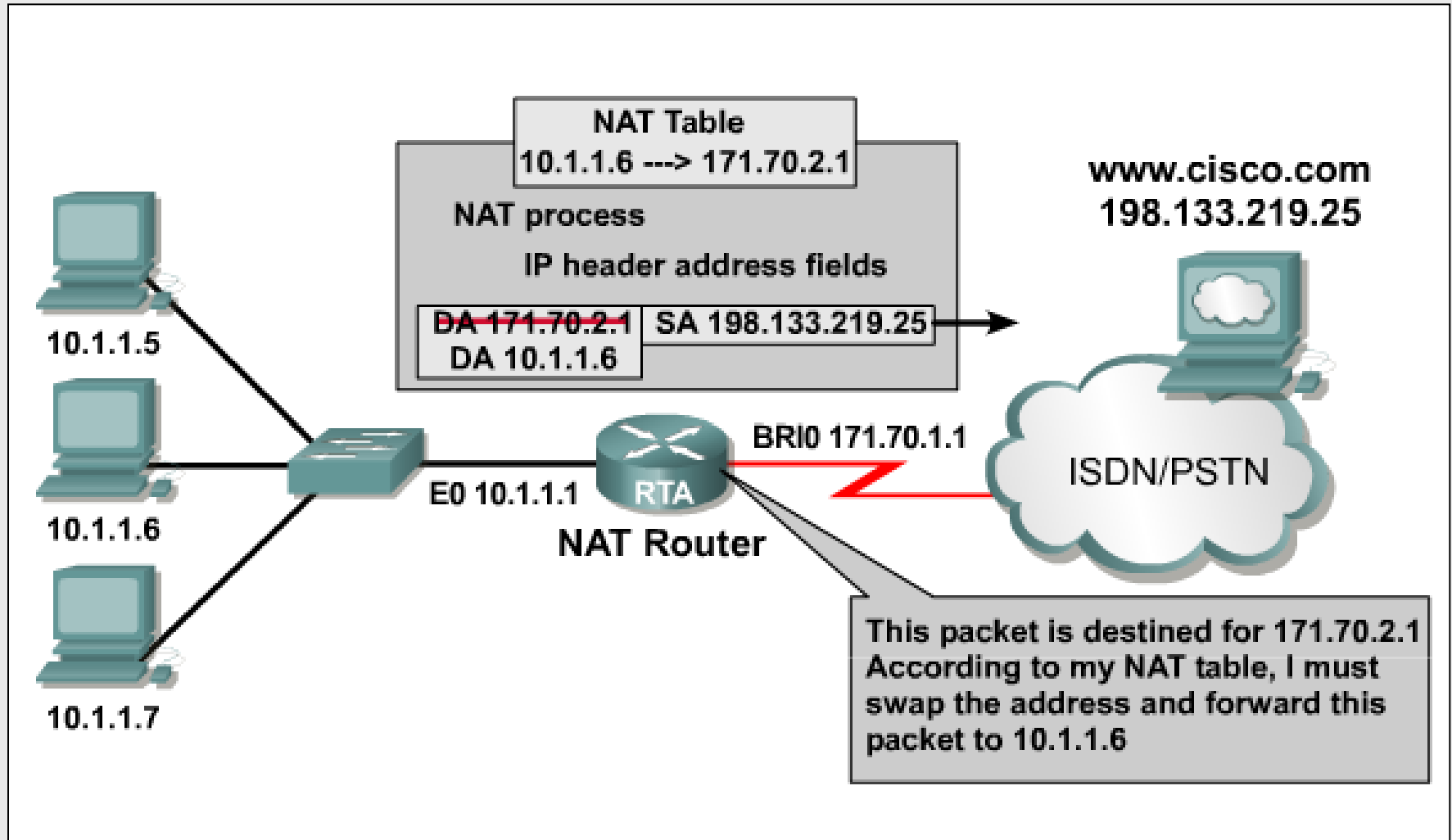
Basic NAT – Example



Basic NAT – Example



Basic NAT – Example



NAT terminology

- **Inside network**

- The network whose addresses are subject to translation

- **Outside network**

- All the other networks whom the inside network is connected to

- **Inside local** IP address

- The configured IP address assigned to a host on the inside network
 - Globally unique
 - Private address
 - Globally unique but not allocated to the organization

- **Inside global** IP address

- The IP address of an inside host as it appears to the outside network

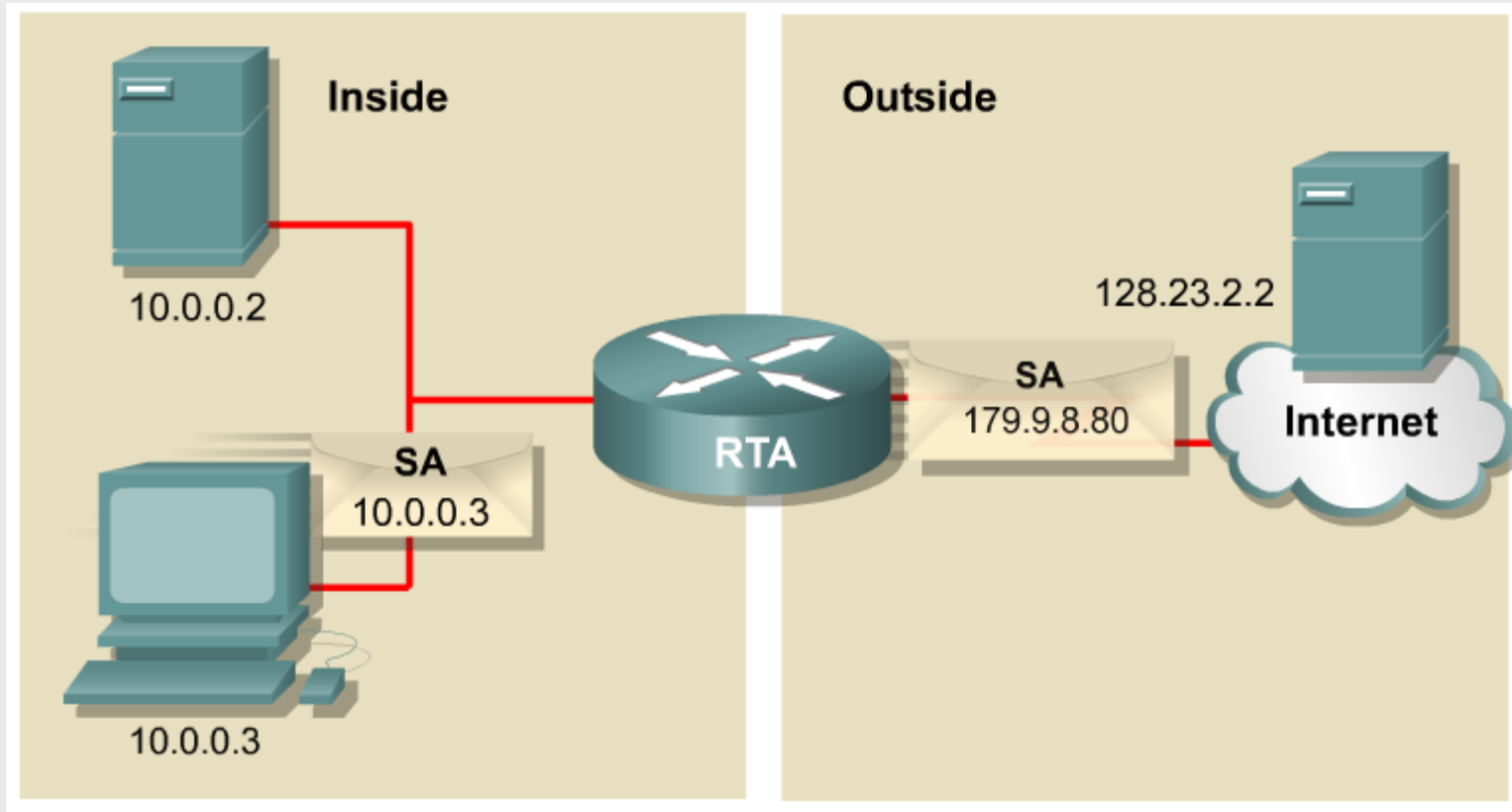
- **Outside local** IP address

- The IP address of an outside host as it appears to the inside network

- **Outside global** IP address

- The configured IP address assigned to a host in the outside network

NAT terminology



NAT Table		
Inside Local IP Address	Inside Global IP Address	Outside Global IP Address
10.0.0.3	179.9.8.80	128.23.2.2
		Outside Local IP Address

Private addresses

- Public IP addresses

- Registered with a Regional Internet Registry (RIR)
- Leased from ISPs

Class	RFC 1918 Internal Address Range	CIDR Prefix
A	10.0.0.0 - 10.255.255.255	10.0.0.0/8
B	172.16.0.0 - 172.31.255.255	172.16.0.0/12
C	192.168.0.0 - 192.168.255.255	192.168.0.0/16

- IANA set aside three blocks for **private IP addresses** [RFC 1918]

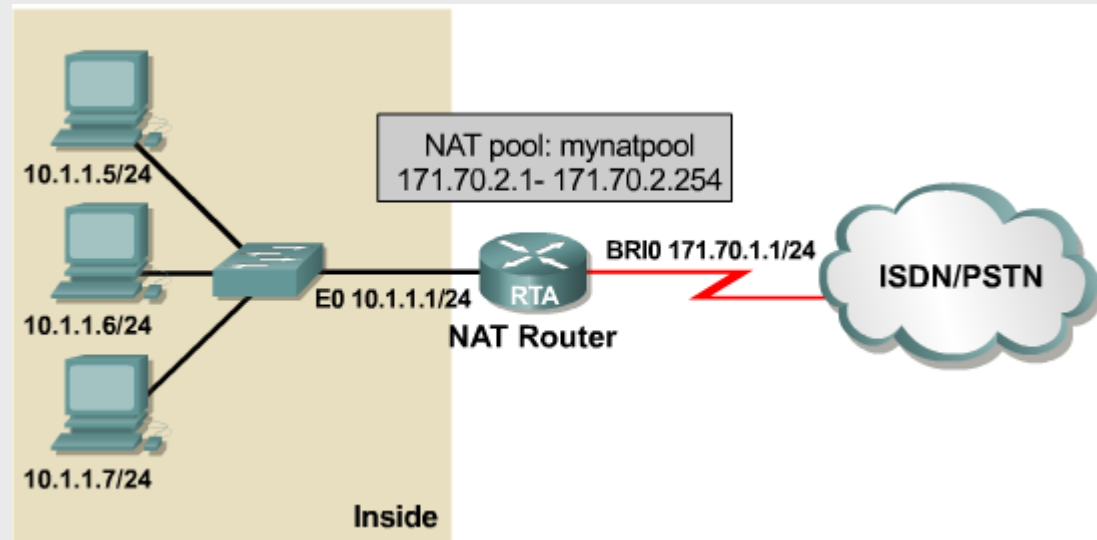
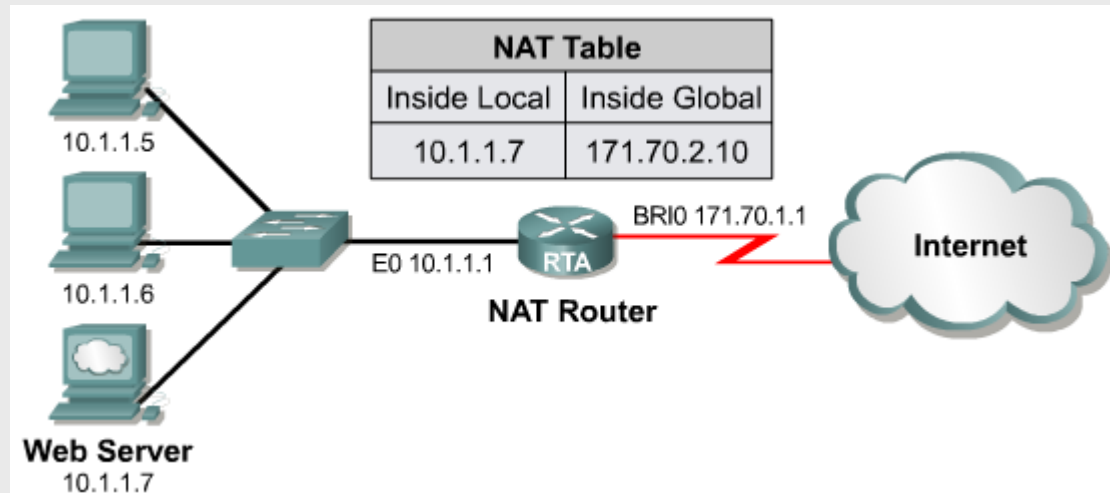
- Use is unrestricted in private networks
- Cannot be routed on the public Internet (ISP routers are configured accordingly)
 - Global uniqueness is not guaranteed

- Advantages and disadvantages of using private address space

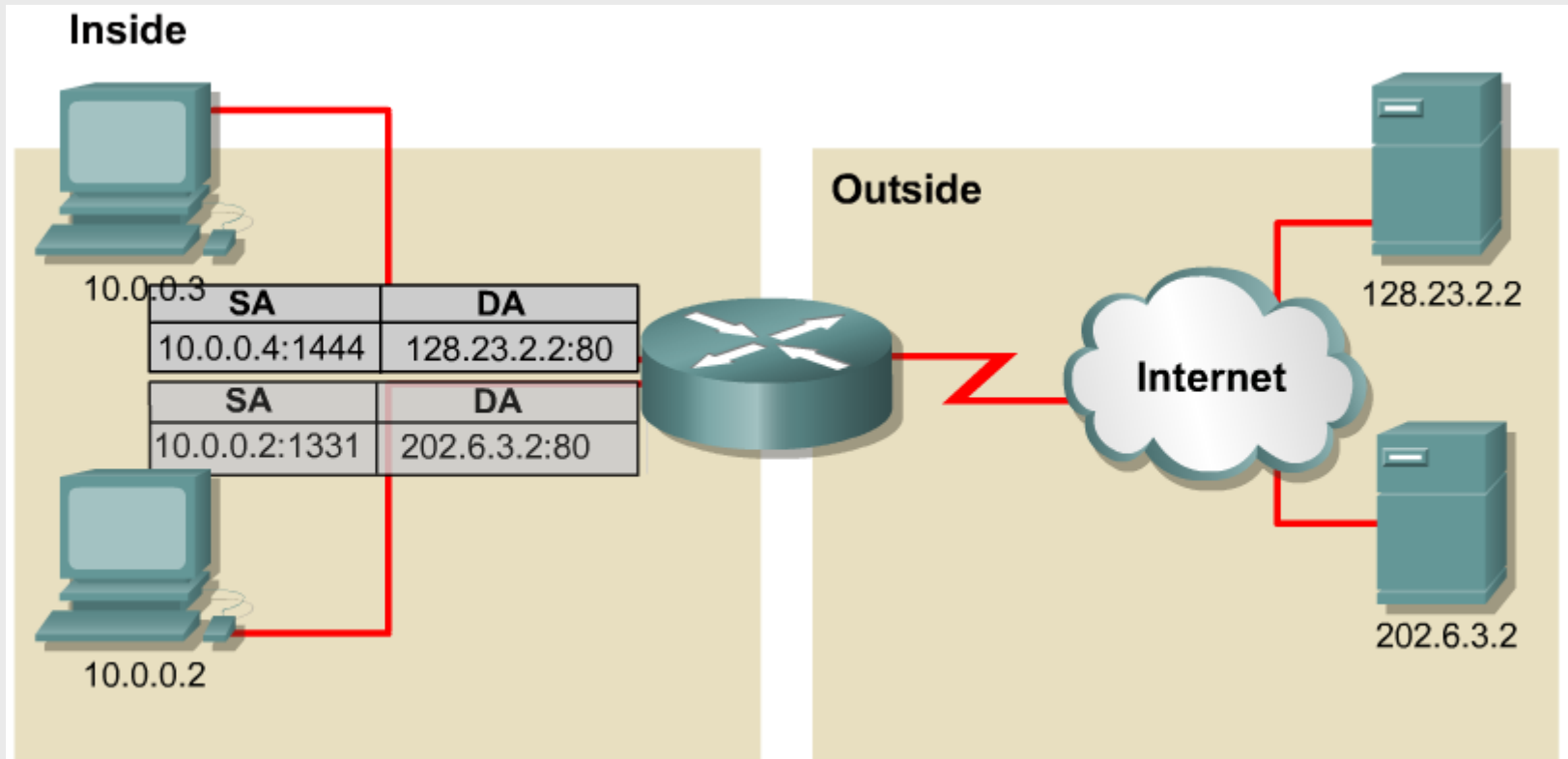
- Flexibility in network design due to larger address space at disposal
- Safe choice once outside connectivity is needed
 - but Network Address Translation is needed then
- May require renumbering in case of several networks merging

Static vs. dynamic NAT

- Pre-configured one-to-one global and local inside address association
- Needed when the inside host (typically a server) shall be accessible from the outside network
- Global and local inside address association is temporarily determined
- Global inside addresses are picked from a pool
 - The pool size is typically smaller than the size of the inside network



NAPT – example



NAT Table			
Inside Local IP Address	Inside Global IP Address	Outside Local IP Address	Outside Global Address
10.0.0.2:1331	179.9.8.20:1331	202.6.3.2:80	202.6.3.2:80
10.0.0.3:1555	179.9.8.20:1555	128.23.2.2:80	128.23.2.2:80

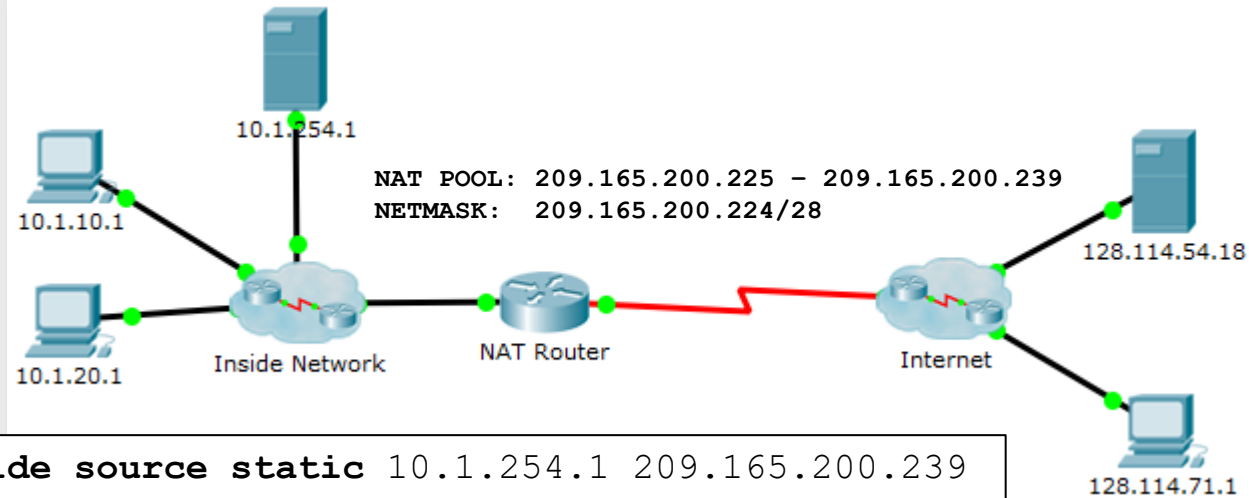
Pros

- Address saving through reuse
- Increase flexibility when connecting to an ISP
 - Inside network size and structure are decoupled from the amount of available public addresses
 - No readdressing is needed when changing the ISP
- Typical use cases
 - Static or dynamic inside network address translation
 - Global address overloading
 - **TCP load balancing**
 - **Overlapping networks**

Cons

- Latency increases
 - Translation adds to forwarding processing
- Loss of end-to-end IP traceability
- Some applications stop working
 - When IP addresses are carried in the packet payload
 - **Application Level Gateways (ALG)** are needed

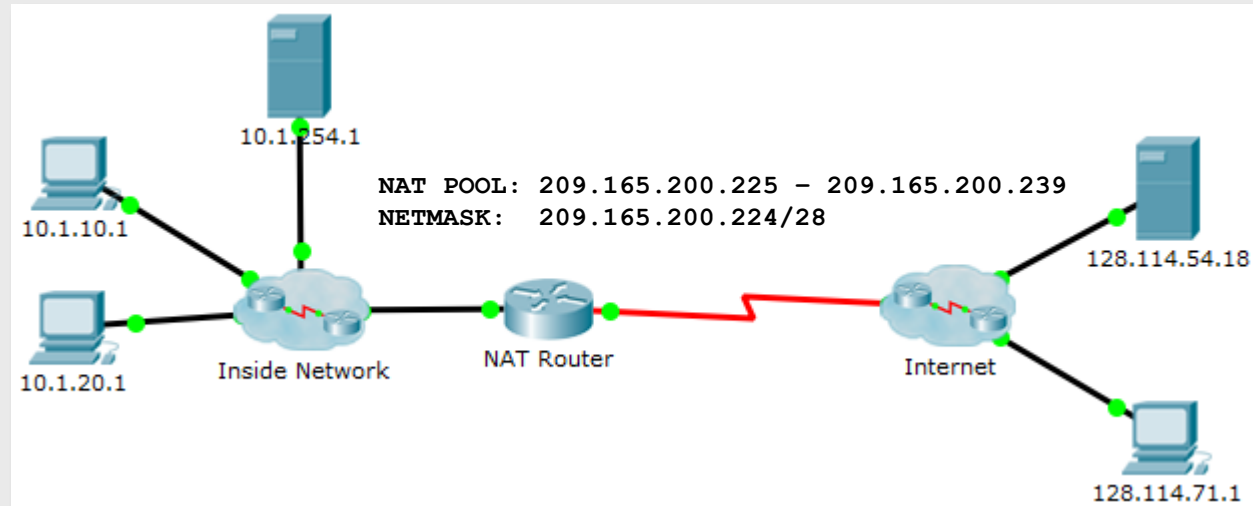
Static NAT configuration



```
Router(config)#ip nat inside source static 10.1.254.1 209.165.200.239
Router(config)#int se0/0/0
Router(config-if)#ip nat outside
Router(config-if)#int fa0/0
Router(config-if)#ip nat inside
```

Step	Action	Notes
1	Establish static translation between an inside local address and an inside global address. Router(config)# ip nat inside source static local-ip global-ip	Enter the global command no ip nat inside source static to remove the static source translation.
2	Specify the inside interface. Router(config)# interface type number	Enter the interface command. The CLI prompt will change from (config)# to (config-if)#.
3	Mark the interface as connected to the inside. Router(config-if)# ip nat inside	
4	Exit interface configuration mode. Router(config-if)# exit	
5	Specify the outside interface. Router(config)# interface type number	
6	Mark the interface as connected to the outside. Router(config-if)# ip nat outside	

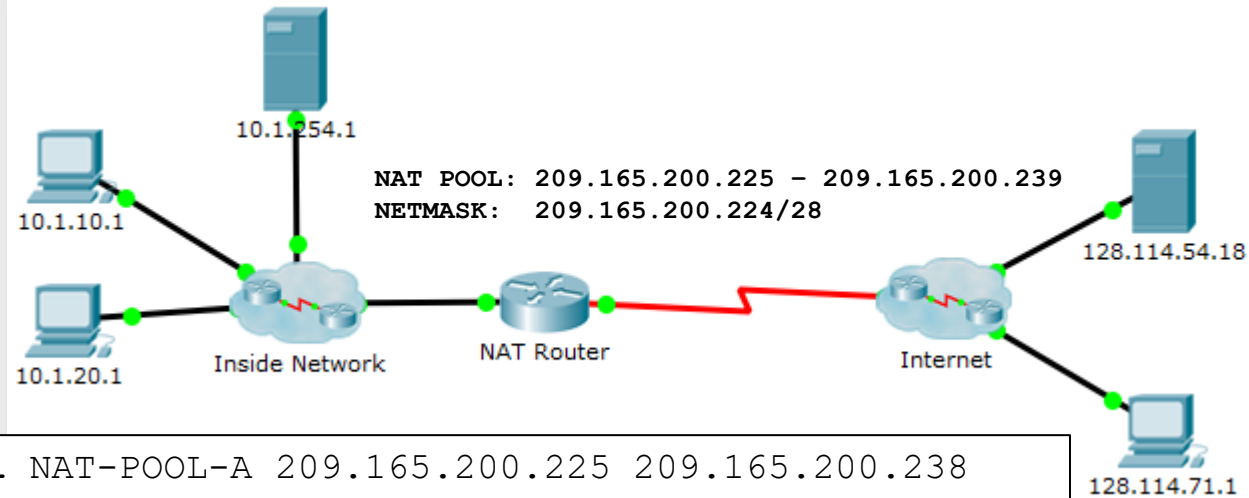
Verifying NAT



```
Router#show ip nat translations
```

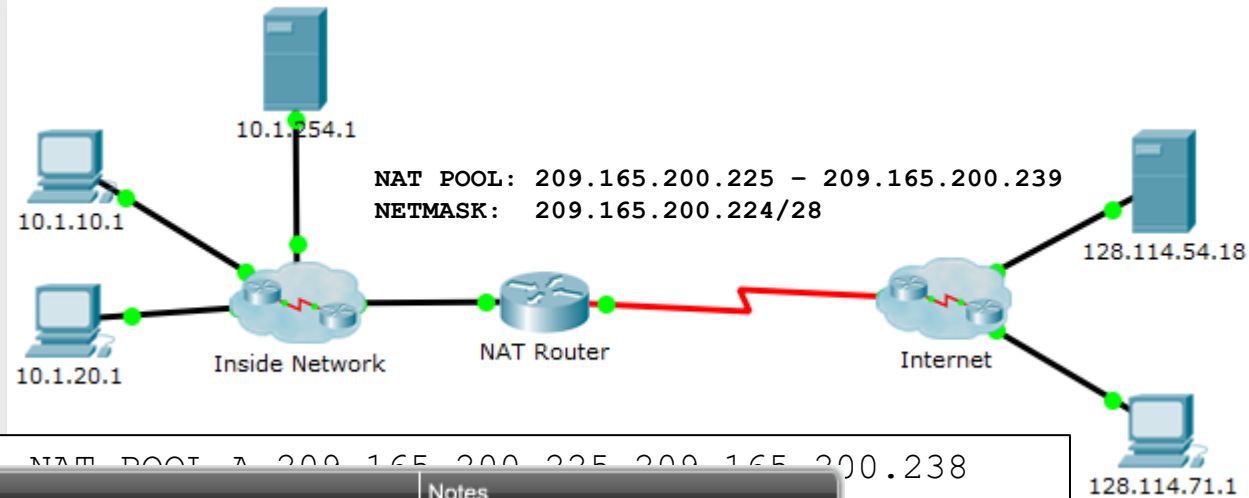
Pro	Inside global	Inside local	Outside local	Outside global
---	209.165.200.239	10.1.254.1	---	---

Dynamic NAT configuration



```
Router(config)#ip nat pool NAT-POOL-A 209.165.200.225 209.165.200.238
netmask 255.255.255.240
Router(config)#ip access-list standard NAT-LIST-INSIDE
Router(config-std-nacl)#permit 10.1.10.0 0.0.0.255
Router(config-std-nacl)#permit 10.1.20.0 0.0.0.255
Router(config-std-nacl)#exit
Router(config)#ip nat inside source list NAT-LIST-INSIDE pool NAT-POOL-A
Router(config)#int se0/0/0
Router(config-if)#ip nat outside
Router(config-if)#int fa0/0
Router(config-if)#ip nat inside
```

Dynamic NAT configuration



```
Router(config)#ip nat pool NAT-POOL-A 209.165.200.225 209.165.200.238
```

```
netmask 255.255.255.0
```

```
Router(config)#
```

```
Router(config)#
```

```
Router(config)#
```

```
Router(config)#
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Router(config)#
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Router(config)#
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Router(config)#
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```
Router(config)#
```

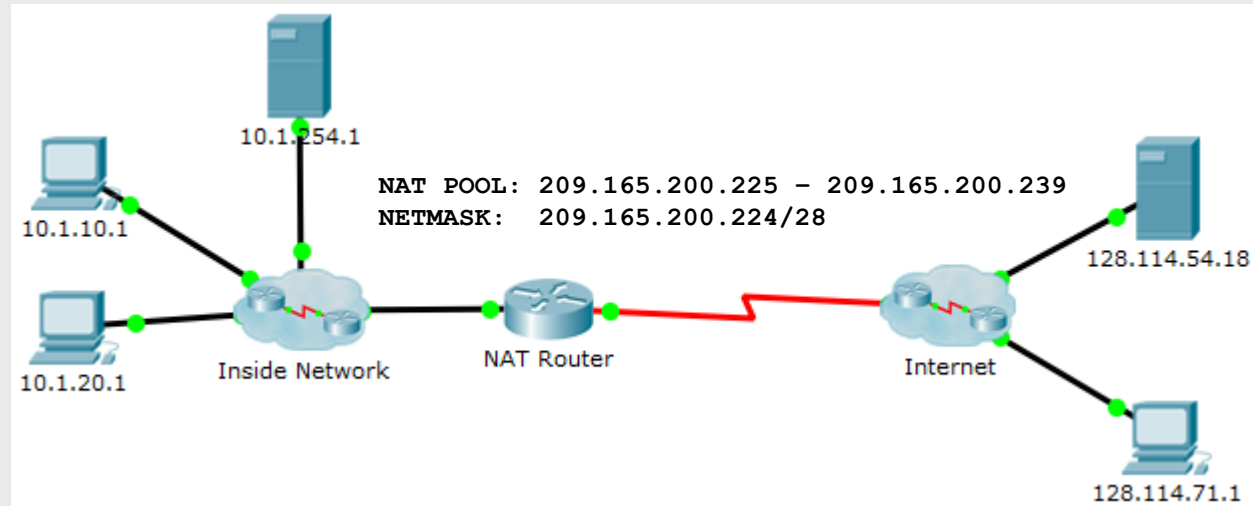
```
Router(config)#
```

```
Router(config)#
```

Step	Action	Notes
1	Define a pool of global addresses to be allocated as needed. Router(config)# ip nat pool name start-ip end-ip { netmask netmask prefix-length prefix-length}	Enter the global command no ip nat pool name to remove the pool of global addresses.
2	Define a standard access list permitting those addresses that are to be translated. Router(config)# access-list access-list-number permit source [source-wildcard]	Enter the global command no access-list access-list-number to remove the access list.
3	Establish dynamic source translation, specifying the access list defined in the prior step. Router(config)# ip nat inside source list access-list-number pool name	Enter the global command no ip nat inside source to remove the dynamic source translation.
4	Specify the inside interface. Router(config)# interface type number	Enter the interface command. The CLI prompt will change from (config)# to (config-if)#.
5	Mark the interface as connected to the inside. Router(config-if)# ip nat inside	
6	Specify the outside interface. Router(config)# interface type number	
7	Mark the interface as connected to the outside. Router(config-if)# ip nat outside	
8	Exit interface configuration mode. Router(config-if)# exit	

NAT-POOL-A

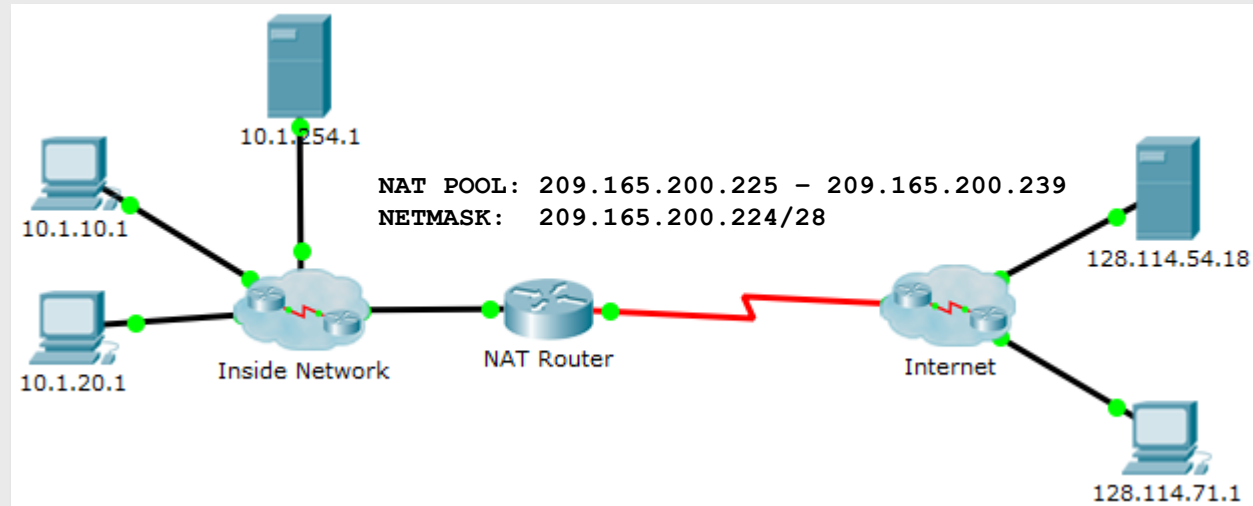
Verifying NAT



```
Router#show ip nat translations
```

Pro	Inside global	Inside local	Outside local	Outside global
icmp	209.165.200.225:1	10.1.10.1:1	128.114.54.18:1	128.114.54.18:1
icmp	209.165.200.225:2	10.1.10.1:2	128.114.54.18:2	128.114.54.18:2
icmp	209.165.200.225:3	10.1.10.1:3	128.114.54.18:3	128.114.54.18:3
icmp	209.165.200.225:4	10.1.10.1:4	128.114.54.18:4	128.114.54.18:4
---	209.165.200.239	10.1.254.1	---	---
tcp	209.165.200.226:1025	10.1.20.1:1025	128.114.54.18:80	128.114.54.18:80

Verifying NAT



```
Router#show ip nat translations
```

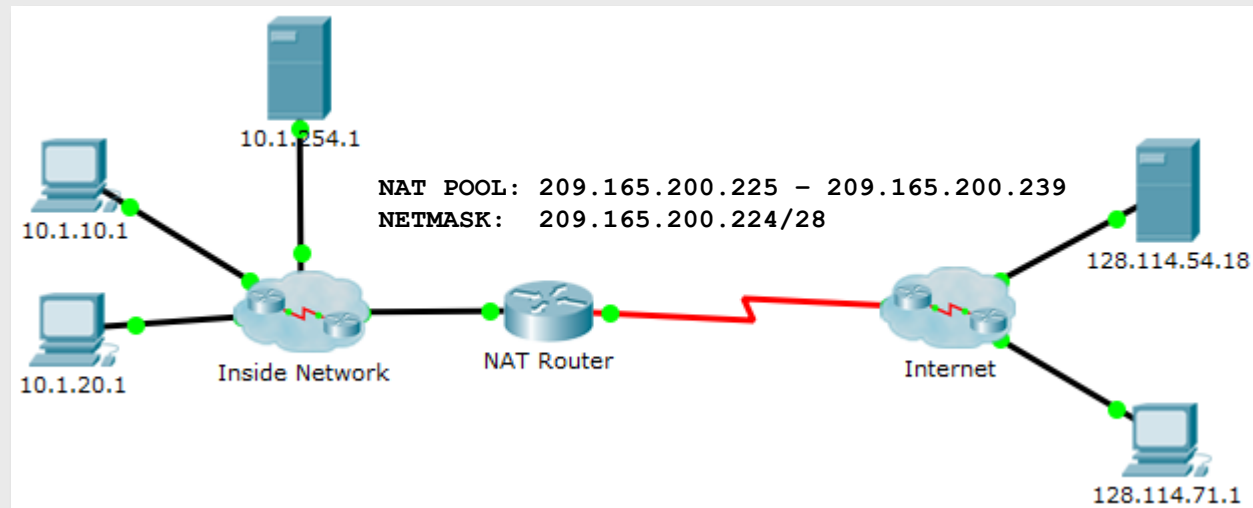
Pro	Inside global	Inside local	Outside local	Outside global
icmp	209.165.200.225:1	10.1.10.1:1	128.114.54.18:1	128.114.54.18:1
icmp	209.165.200.225:2	10.1.10.1:2	128.114.54.18:2	128.114.54.18:2
icmp	209.165.200.225:3	10.1.10.1:3	128.114.54.18:3	128.114.54.18:3
icmp	209.165.200.225:4	10.1.10.1:4	128.114.54.18:4	128.114.54.18:4
---	209.165.200.239	10.1.254.1	---	---
tcp	209.165.200.226:1025	10.1.20.1:1025	128.114.54.18:80	128.114.54.18:80

```
Router#clear ip nat translations *
```

```
Router#show ip nat translations
```

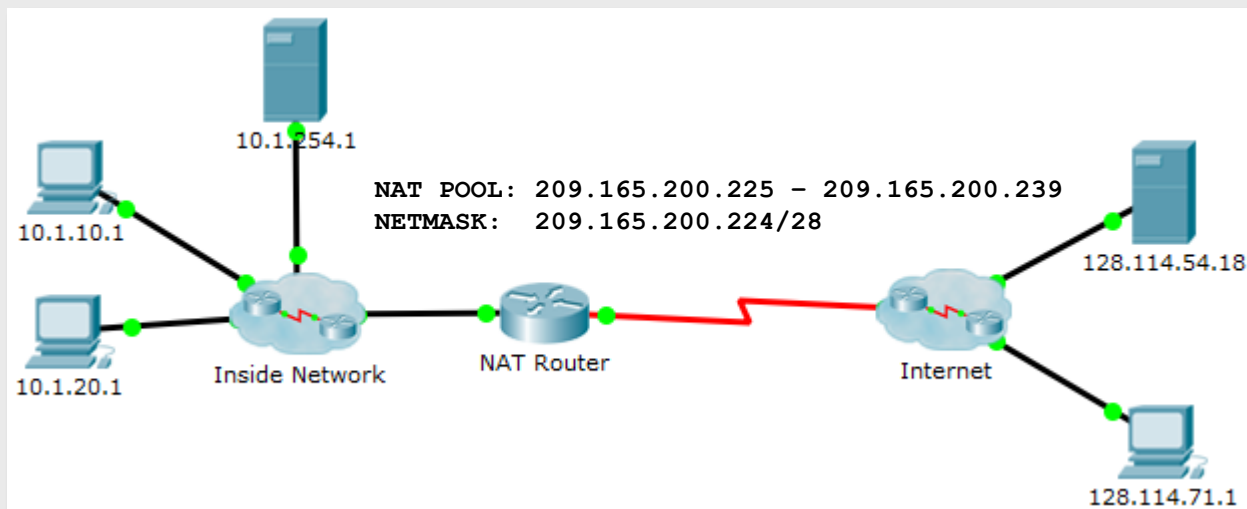
Pro	Inside global	Inside local	Outside local	Outside global
---	209.165.200.239	10.1.254.1	---	---

NAT overload configuration (pool)



```
Router(config)#ip nat pool NAT-POOL-A 209.165.200.225 209.165.200.238 netmask
255.255.255.240
Router(config)#ip access-list standard NAT-LIST-INSIDE
Router(config-std-nacl)#permit 10.1.10.0 0.0.0.255
Router(config-std-nacl)#permit 10.1.20.0 0.0.0.255
Router(config-std-nacl)#exit
Router(config)#ip nat inside source list NAT-LIST-INSIDE pool NAT-POOL-A overload
Router(config)#int se0/0/0
Router(config-if)#ip nat outside
Router(config-if)#int fa0/0
Router(config-if)#ip nat inside
```

NAT overload configuration (pool)



```
Router(config)#ip nat pool NAT-POOL-A 209.165.200.225 209.165.200.238 netmask  
255.255.255.24
```

```
Router(config)
```

```
Router(config-
```

```
Router(config-
```

```
Router(config-
```

```
Router(config)
```

```
Router(config)
```

```
Router(config-
```

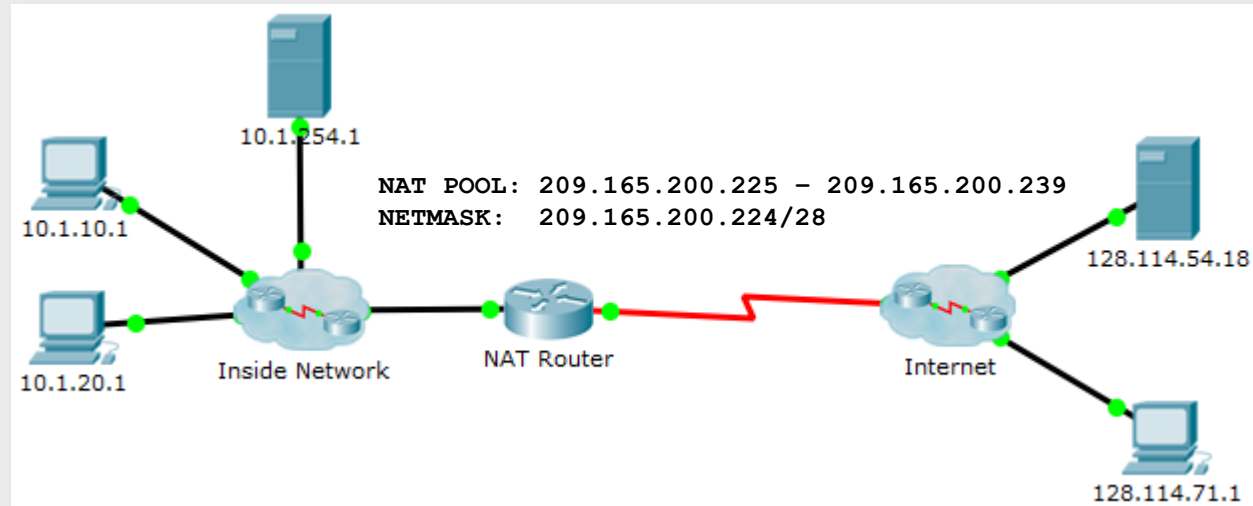
```
Router(config-
```

```
Router(config-
```

Step	Action	Notes
1	Define a standard access list permitting those addresses that are to be translated. Router(config)# access-list acl-number permit source [source-wildcard]	Enter the global command no access-list access-list-number to remove the access list.
2	Specify the global address, as a pool, to be used for overloading. Router(config)# ip nat pool name start-ip end-ip { netmask netmask prefix-length prefix-length}	
3	Establish overload translation. Router {config} # ip nat inside source list acl-number pool name overload	
4	Specify the inside interface. Router(config)# interface type number Router(config-if)# ip nat inside	Enter the interface command. The CLI prompt will change from (config)# to (config-if)#.
5	Specify the outside interface. Router(config-if)# interface type number Router(config-if)# ip nat outside	

OL-A **overload**

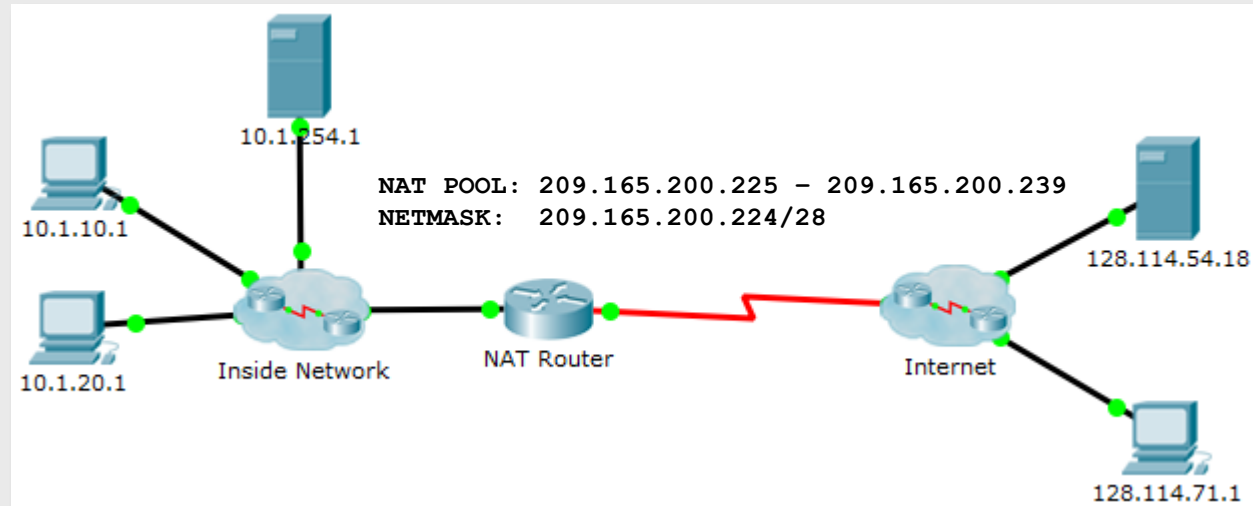
Verifying NAT



```
Router#show ip nat translations
```

Pro	Inside global	Inside local	Outside local	Outside global
---	209.165.200.239	10.1.254.1	---	---
tcp	209.165.200.225:1026	10.1.10.1:1026	128.114.54.18:80	128.114.54.18:80
tcp	209.165.200.225:1028	10.1.20.1:1028	128.114.54.18:80	128.114.54.18:80

Verifying NAT



```
Router#show ip nat statistics
```

```
Total translations: 7 (1 static, 6 dynamic, 6 extended)
```

```
Outside Interfaces: Serial0/0/0
```

```
Inside Interfaces: FastEthernet0/0
```

```
Hits: 148 Misses: 56
```

```
Expired translations: 28
```

```
Dynamic mappings:
```

```
-- Inside Source
```

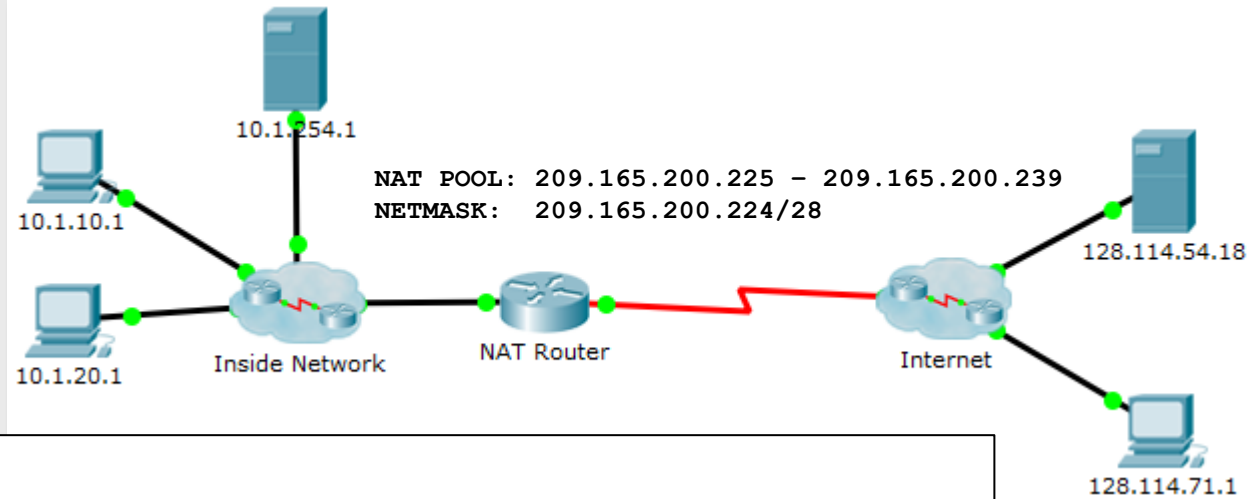
```
access-list NAT-LIST-INSIDE pool NAT-POOL-A refCount 6
```

```
pool NAT-POOL-A: netmask 255.255.255.240
```

```
start 209.165.200.225 end 209.165.200.238
```

```
type generic, total addresses 14 , allocated 1 (7%), misses 0
```


Verifying NAT



```
Router#debug ip nat
```

```
IP NAT debugging is on
```

```
Router#
```

```
NAT: s=10.1.20.1->209.165.200.225, d=128.114.54.18 [55]
```

```
NAT*: s=128.114.54.18, d=209.165.200.225->10.1.20.1 [49]
```

```
NAT*: s=10.1.20.1->209.165.200.225, d=128.114.54.18 [56]
```

```
NAT*: s=10.1.20.1->209.165.200.225, d=128.114.54.18 [57]
```

```
NAT*: s=128.114.54.18, d=209.165.200.225->10.1.20.1 [50]
```

```
NAT*: s=10.1.20.1->209.165.200.225, d=128.114.54.18 [58]
```

```
NAT*: s=128.114.54.18, d=209.165.200.225->10.1.20.1 [51]
```

```
NAT*: s=10.1.20.1->209.165.200.225, d=128.114.54.18 [59]
```

```
NAT: s=10.1.10.1->209.165.200.225, d=128.114.54.18 [36]
```

```
NAT*: s=128.114.54.18, d=209.165.200.225->10.1.10.1 [52]
```

```
NAT: s=10.1.10.1->209.165.200.225, d=128.114.54.18 [37]
```

```
NAT*: s=128.114.54.18, d=209.165.200.225->10.1.10.1 [53]
```

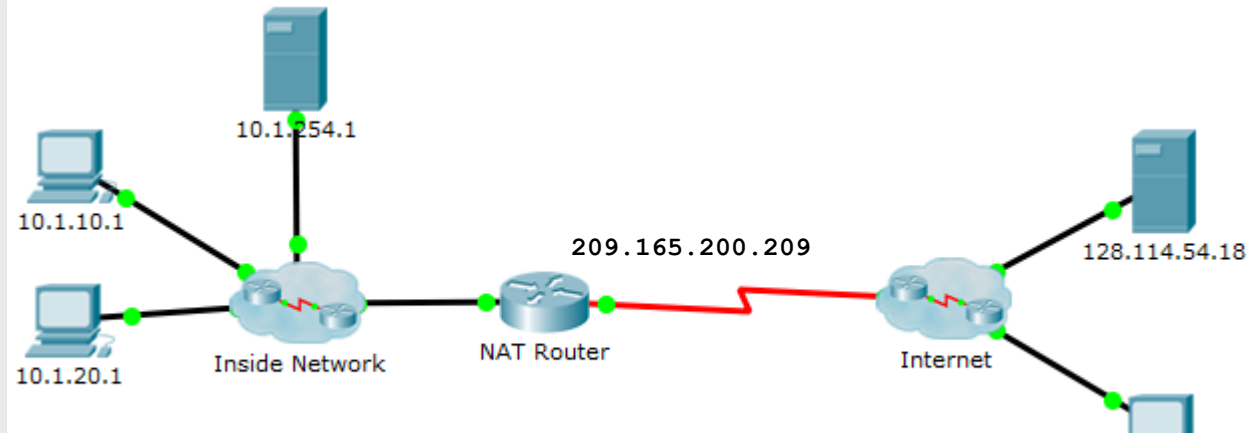
```
NAT: s=10.1.10.1->209.165.200.225, d=128.114.54.18 [38]
```

```
NAT*: s=128.114.54.18, d=209.165.200.225->10.1.10.1 [54]
```

```
NAT: s=10.1.10.1->209.165.200.225, d=128.114.54.18 [39]
```

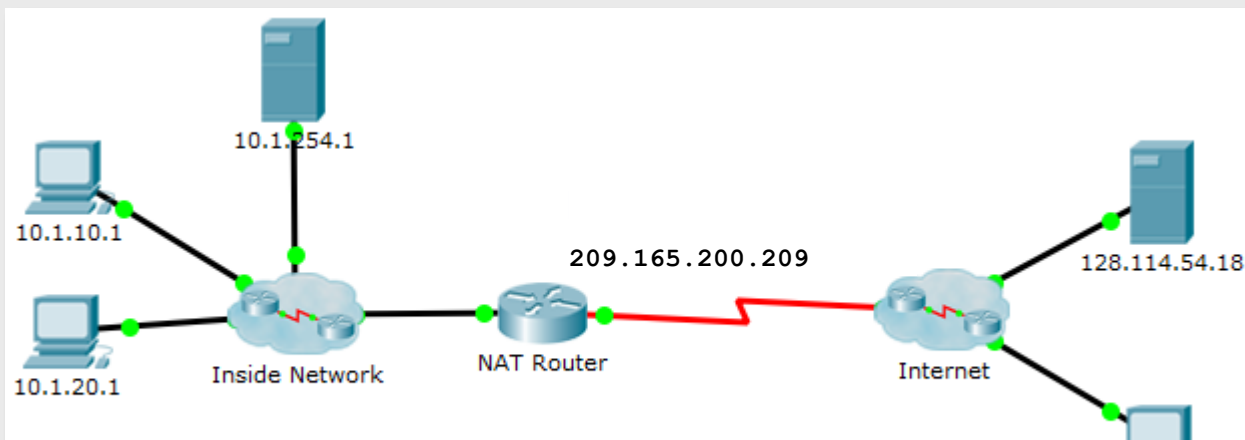
```
NAT*: s=128.114.54.18, d=209.165.200.225->10.1.10.1 [55]
```

NAT overload configuration (one address)



```
Router(config)#ip access-list standard NAT-LIST-INSIDE
Router(config-std-nacl)#permit 10.1.10.0 0.0.0.255
Router(config-std-nacl)#permit 10.1.20.0 0.0.0.255
Router(config-std-nacl)#exit
Router(config)#ip nat inside source list NAT-LIST-INSIDE interface serial 0/0/0
overload
Router(config)#int se0/0/0
Router(config-if)#ip nat outside
Router(config-if)#int fa0/0
Router(config-if)#ip nat inside
```

NAT overload configuration (one address)

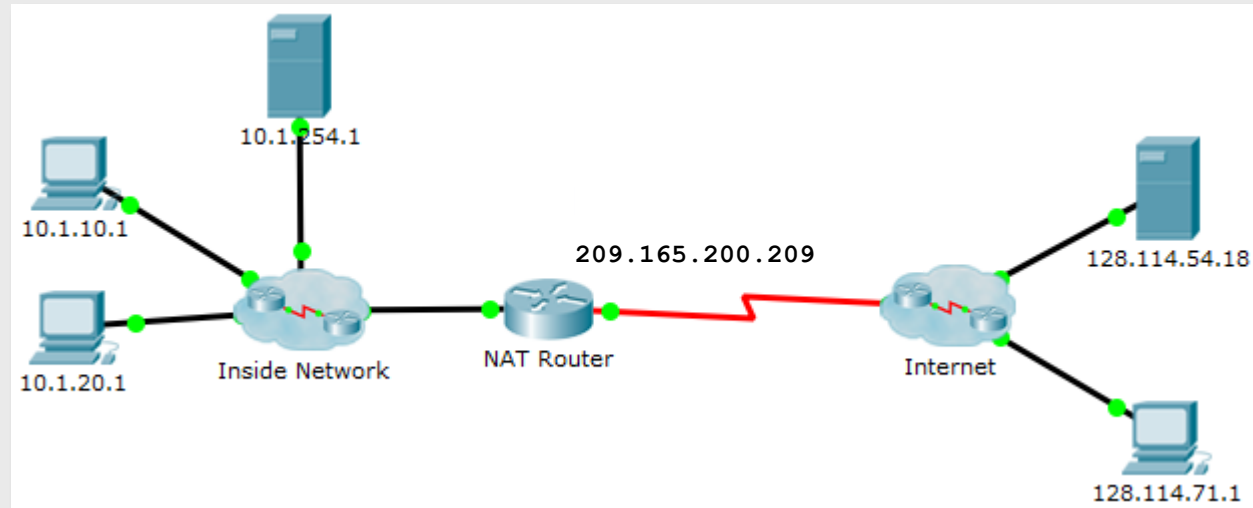


```
Router(config)#ip access-list standard NAT-LIST-INSIDE
Router(config-std-nacl)#permit 10.1.10.0 0.0.0.255
Router(config-std-nacl)#permit 10.1.20.0 0.0.0.255
Router(config-std-nacl)#exit
Router(config)#ip nat inside source list NAT-LIST-INSIDE interface serial 0/0/0
overload
```

```
Router(config)
Router(config)
Router(config)
Router(config)
```

Step	Action	Notes
1	Define a standard access list permitting those addresses that are to be translated. Router(config)# access-list <i>acl-number</i> permit <i>source</i> [<i>source-wildcard</i>]	Enter the global command no access-list <i>access-list-number</i> to remove the access list.
2	Establish dynamic source translation, specifying the access list defined in the prior step. Router(config)# ip nat inside source list <i>acl-number</i> interface <i>interface</i> overload	Enter the global command no ip nat inside source to remove the dynamic source translation. The overload keyword enables PAT.
3	Specify the inside interface. Router(config)# interface <i>type number</i> Router(config-if)# ip nat inside	Enter the interface command. The CLI prompt will change from (config)# to (config-if)#.
4	Specify the outside interface. Router(config-if)# interface <i>type number</i> Router(config-if)# ip nat outside	

Verifying NAT

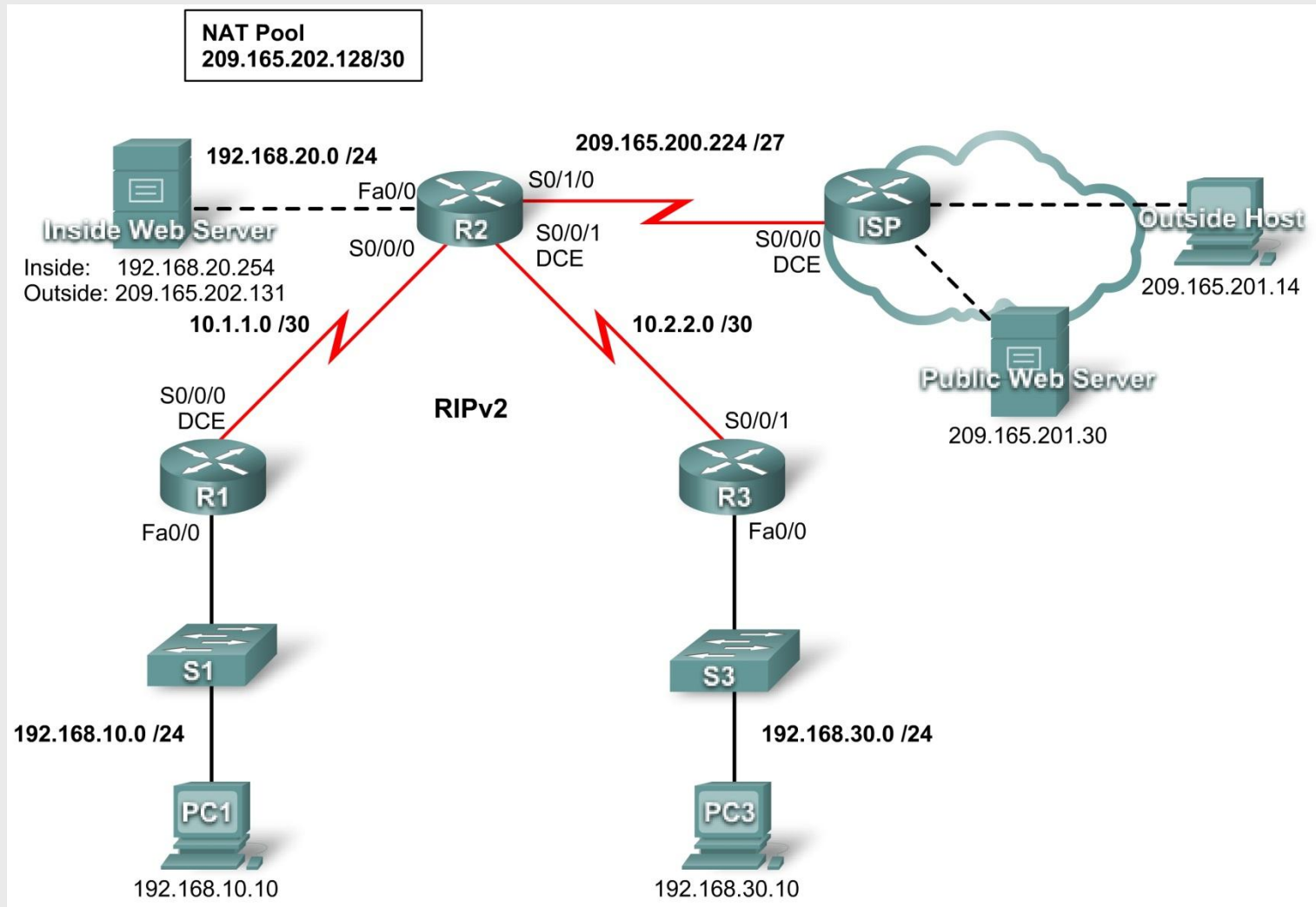


```
Router#show ip nat translations
```

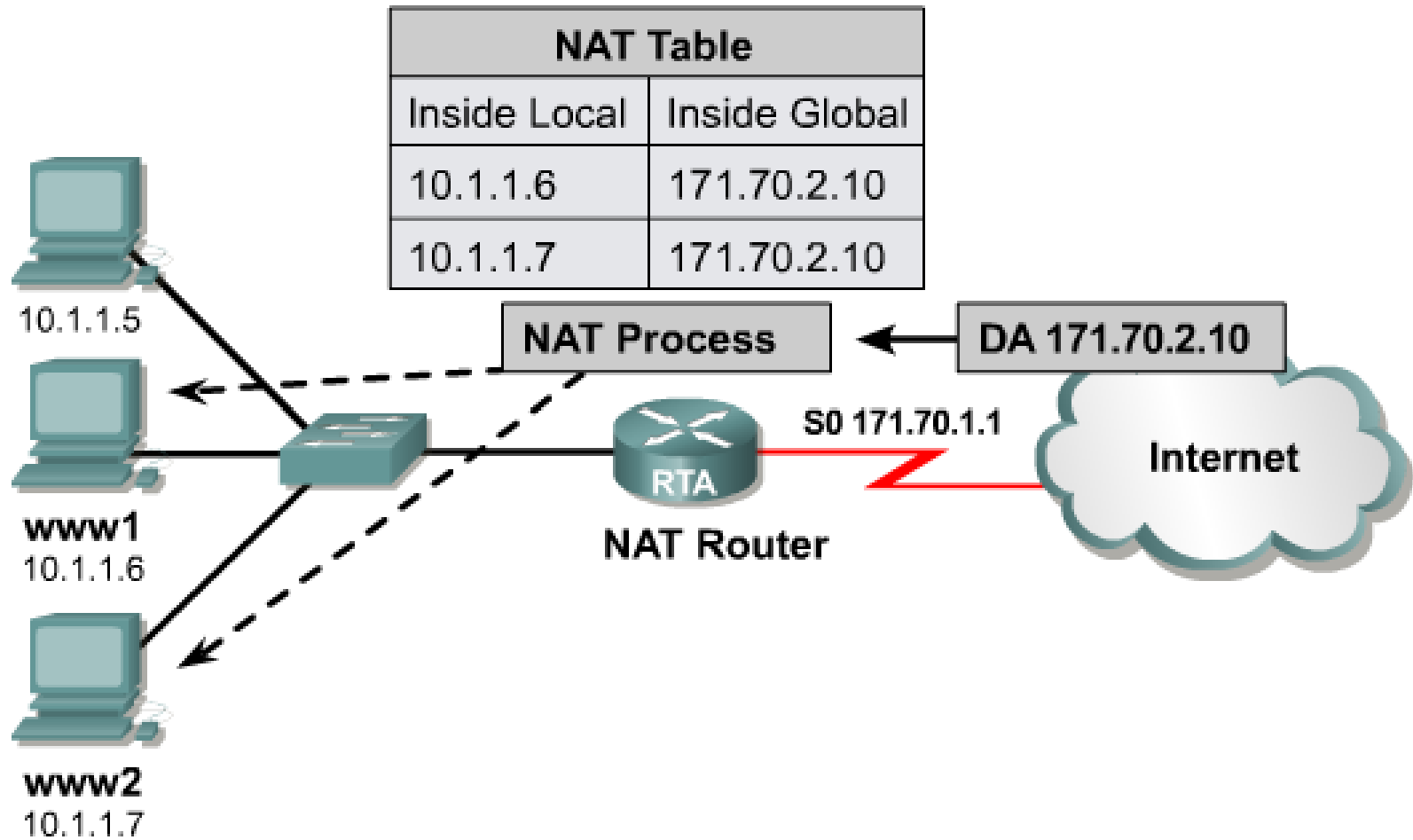
Pro	Inside global	Inside local	Outside local	Outside global
---	209.165.200.239	10.1.254.1	---	---
tcp	209.165.200.209:1029	10.1.10.1:1029	128.114.54.18:80	128.114.54.18:80
tcp	209.165.200.209:1030	10.1.20.1:1030	128.114.54.18:80	128.114.54.18:80

Lab activity

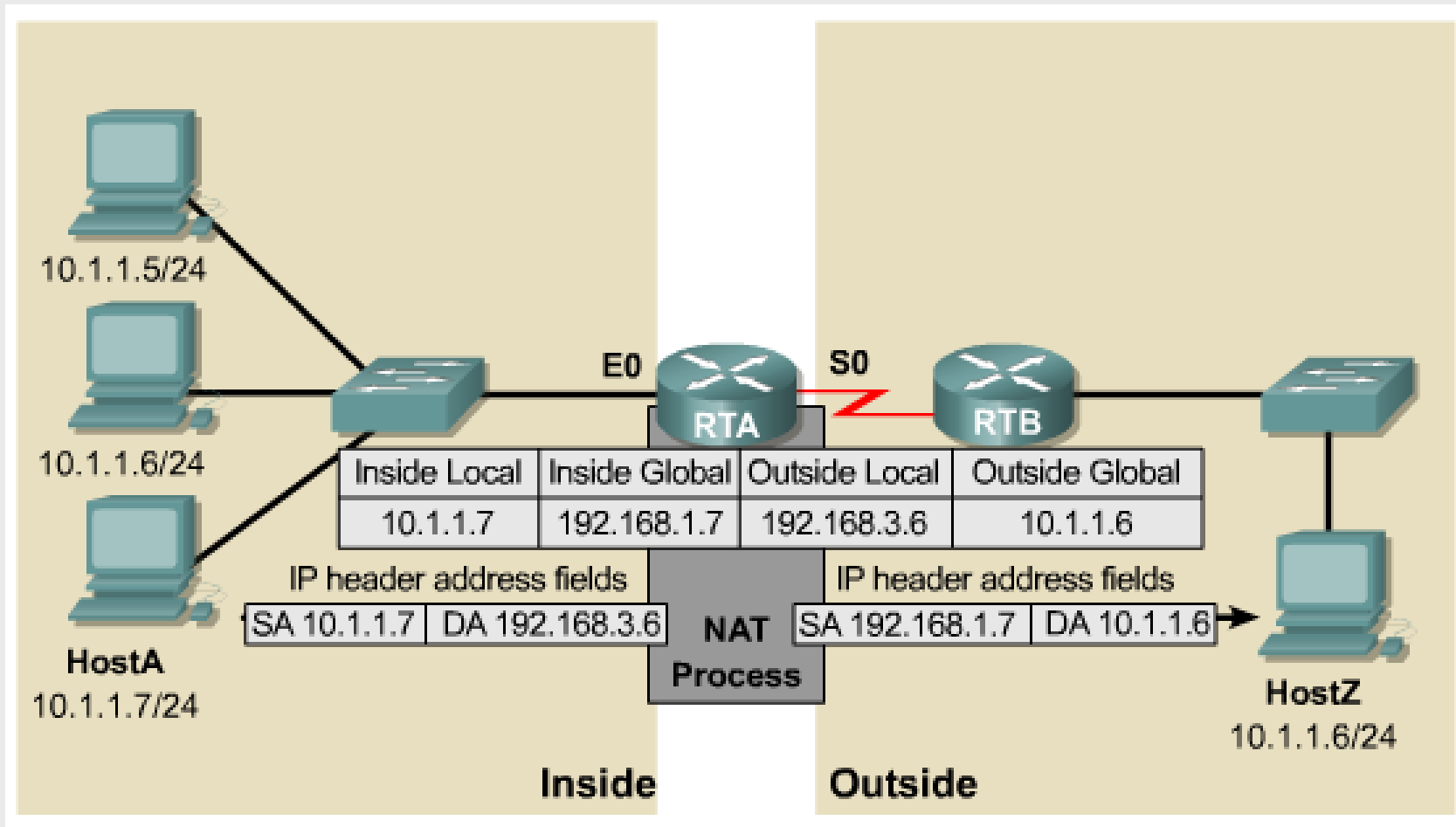
■ NAT configuration



TCP load distribution

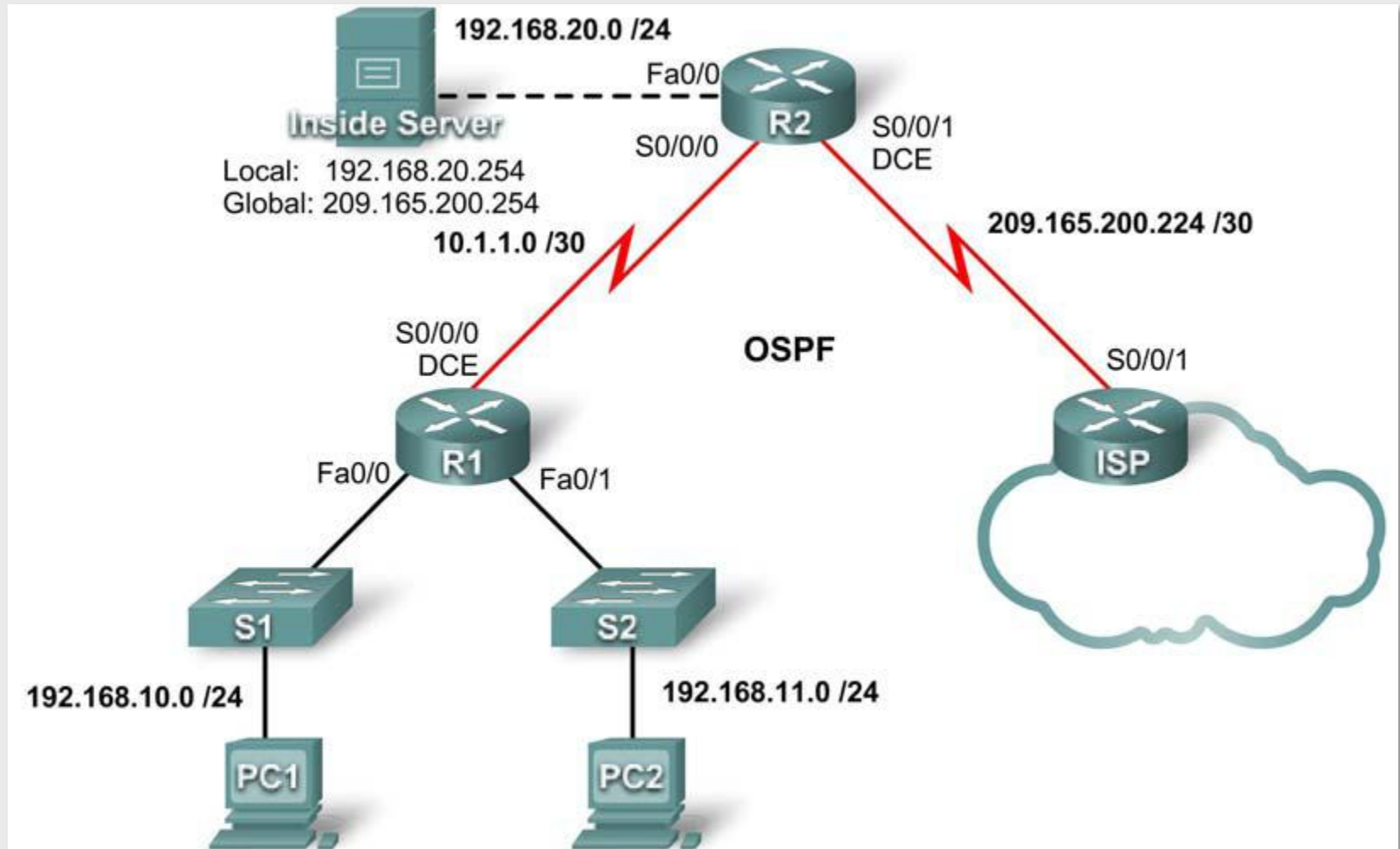


Overlapping networks



Lab activity

■ DHCP and NAT configuration



Lab activity

■ DHCP and NAT configuration

